find the problem areas and focus on innovation to solve them. For the DFMA analysis, the chart given in Table 2 is a valuable tool.

We start our analysis by listing down all the parts that go into the assembly. We first evaluate the essential parts and identify the possibility of reducing the parts. If we are better than 60% in TPI and better than 80% in SPI, we can be satisfied and move to other areas.

Next, we should look at the assembly complexities. A high ACI means we shall face difficulties during assembly and have more quality problems in assembled products. High ACI will also require us to make parts with close tolerances. Normally, if we are within 120% of the target ACI we can say that design is good.

Once we have done the assembly complexity analysis, we should evaluate other process indices. To analyse handling, error proofing, insertion, and secondary operation problems, we check each part against the checklist and mark a tick or 'y' for cases where there is a problem. Finally, we calculate the percentage of such cases against theoretical minimum parts.

For the areas where we have problems in more than 5% of parts, we must review the assembly process and design of the parts to bring the problem cases down.

Design guidelines for assembly

Always thoroughly examine the need for designing a new part. In most cases, we can design products using readily available standard parts. If standard parts do not serve our requirements, we should examine the possibilities of using parts that are already designed for some other product.

Reusing parts helps us to avoid additional manufacturing setup as well as reduces the time required to develop a new part. Sometimes, it may be possible to reuse an existing design with slight modifications.

For ease of assembly, the product must have a base part where all other parts will fit. In cases where multiple such parts compete for the base part, we can think of making sub-assemblies separately and then assembling the main product.

We must remember and distinguish between essential parts and non-essential parts. Parts like fasteners, spacers, washers, O-rings, connectors, leads, pipes, and circlips are not what a customer sees as a product. They may be important for technical reasons but they have no value for the customer. Try to eliminate these non-essential parts.

Of the non-essential parts, a screw is probably the most troublesome one. Screws are a century-old technology. They are difficult to locate and difficult to fix. After repeated use, screw threads wear out and become a maintenance problem.

A study by Ford Motor Co. revealed that threaded fasteners were the most common cause of warranty repairs. A similar finding has been echoed in other surveys as well. Use screws in a product only as a last resort. If you are fastening or gluing some parts, then examine the possibility of combining them.

Cost-wise, screwing is the costliest option for fastening, followed by riveting, plastic bending, and snap-fit in decreasing order. If we must fasten two components, examine the possibility to use snap-fitted parts.

Symmetrical parts are always oriented properly. If the product requires a part to be asymmetrical, make sure that part can only be fitted with correct orientation.

Gravity helps to keep the parts in place. If we ensure that parts can be stacked from the top, gravity becomes our friend, and during assembly the parts will remain in place when we place them on top of another part. If we try to assemble anything from the bottom, it becomes a very difficult operation.

Provide features to guide the parts to be assembled into their correct place. Features like counter sinking, recess, and projection help to guide components in their correct location.

Consider handling difficulties during assembly. Closing spring ends, using flanges, and avoiding sharp locking angles ensure that parts will nest or tangle in the storage bins.

Try to avoid restricted access during assembly. Try to keep the base part, where most other parts will be assembled, open from all sides. This will help free access to tools during assembly. Ensure that operators will be able to see the parts clearly during the assembly process.

To summarise:

- Minimise part count
- Design parts that are self-locating
- Design parts that are self-fastening
- Minimise reorientation of parts during assembly
- Design for component symmetry to minimise reorientation
- Design parts for easy retrieval, handling, and insertion
- Plan for fault-proof assembly
- Use drop-down assembly that makes gravity a friend
- Standardise parts and reuse parts already designed
- Encourage modular design
- Plan for a base part to locate other components

By embracing simplicity and efficiency in our designs, considering manufacturing operations, and utilising measurement techniques like ElMaraghy complexity index, we can unlock the potential for improved manufacturability, reduced costs, and successful product realisation. EFY